MOVIE METER

AI-Powered IMDb Rating Category Predictor & Success Analytics Platform

Document Type: Technical Architecture & Design Document

Target Domain: South Indian Cinema & OTT Distribution Strategy

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1. Executive Summary & Problem Statement

In the commercial film industry, particularly within South Indian regional cinema, pre-production planning and budget allocations are frequently based on subjective intuition. Over-investment or improper distribution negotiations can lead to severe capital losses for production houses and OTT networks.

Movie Meter is an enterprise-grade machine learning platform designed to predict a film's quality category (High, Medium, or Low) using pre-release metadata (genres, runtime, language, content certification, and target casting reputation metrics). By mapping superstar and director popularity coefficients in a leak-free out-of-fold manner, the platform provides distribution matching, revenue estimation ranges, and demographic projections before theatrical release.

2. Technical Pipeline & Feature Engineering

The model is trained on a merged and cleaned database of 4,900+ films. The pipeline strictly excludes post-release parameters (such as actual ticketing revenues, review texts, or voter counts) to prevent data leakage.

2.1 Reputation Mappings & Target Encoding

High-cardinality nominal parameters like *director_name* and *actor_1_name* pose severe challenges to traditional one-hot encoders. To resolve this, our preprocessor computes cross-validated out-of-fold target averages for IMDb scores. The average rating achieved by a director or actor's previous works serves as a continuous numerical score mapping. For new or fictional projects (e.g., Nelson Dilipkumar or Superstar Vijay), historical weights are calculated on startup to simulate prediction outcomes accurately.

2.2 Preprocessing Details

Runtimes are parsed and cleaned of empty fields using median values, while content certification labels are bucketed into clean categories (e.g. PG-13, R, G). High skew features are scaled using standard normalization to prevent gradient boosting scaling biases.

3. Model Selection & Hyperparameter Tuning

A rigorous model selection process compared four primary algorithms using 5-Fold Stratified Cross-Validation to assess classification boundaries:

Model Architecture	CV Accuracy	ROC-AUC	LogLoss	Status
Extreme Gradient Boosting (XGBoost)	62.1%	73.0%	0.85	Best / Selected
Random Forest Classifier	60.4%	71.2%	0.89	Baseline
Gradient Boosting Machine (GBM)	59.8%	70.1%	0.91	Baseline
Logistic Regression (Baseline)	54.2%	64.0%	1.05	Discarded

4. Streamlit SaaS Redesign & Architecture

The web application frontend has been redesigned to align with premium SaaS platforms (Vercel, Apple, Stripe):

• **Single-Page Smooth Scrolling:** Replaced the heavy tabbed structure with a single-page scrolling layout where sections are stacked vertically. This guarantees fluid transitions and prevents layout lag.

• **Sticky Anchor Links Navigation:** A custom navigation bar remains fixed at the top of the window, providing responsive shortcut buttons that smooth-scroll to specific sections.

• **High Contrast Input Rendering:** Overrode all Streamlit widget labels to render in bold dark slate (#0f172a). Input textboxes and select options are styled in clean white panels to prevent light-theme rendering bugs on browser dark-mode defaults.

• **Plotly Success Analytics:** Integrates dynamic, responsive gauges showing success probabilities, market potentials, and production readiness index models.

5. Revenue Projections & Distribution Strategy

Understanding rating predictions in isolation is insufficient for commercial stakeholders. The platform translates prediction results into financial projections and release recommendations:

5.1 Gross Box Office Projection Model

Using starcast reputation averages and target genres as multiplier coefficients, gross revenues are projected across international (USD) and regional South Indian (INR Crores) markets. These estimates are mapped against standard industry budgets to define a projected ROI percentage boundaries window.

5.2 Digital OTT Licensing Windows

Based on classifications (High/Medium/Low success potential), the model recommends optimal post-theatrical streaming platforms (e.g. Netflix for high crossover titles; Hotstar for regional action; Sun NXT for indie/low-budget titles) and lists guidelines for release scheduling.

***Operational Guideline:** The theatrical exclusive window must be maintained at a minimum of 4-6 weeks for South Indian theater chains to maximize local box office yields, before shifting to digital streaming platforms.*